LLM-6XX | Assignment 1

Urdu News Headline Classification with Parameter-Efficient Fine-Tuning

Stage 1 Track B Delivery Report
Infinitic AI coursework project

Group members

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Finalist model

xlm-roberta-base · full fine-tune · cosine schedule

Primary results

Validation macro-F1: **0.9481** | Test macro-F1: **0.9384**

Deadline: 20 October 2026

Repository: github.com/HammadHARahim/Infinitic-AI-Urdu-News-Headline-Classifier

1. Project description

Given an Urdu news headline, predict its news category. Urdu is a low-resource language written in a right-to-left Perso-Arabic script. Tokenizers built mostly on English often split Urdu words into many subword pieces. This project measures how well a multilingual pretrained transformer handles that setting, how close LoRA gets to full fine-tuning, and which regularization and schedule choices improve validation macro-F1.

Research questions addressed in Stage 1

- RQ1: How close does LoRA get to full fine-tuning, and with how many trainable parameters?
- RQ2 (partial): How does the XLM-R tokenizer behave on Urdu headlines?
- RQ4 (partial): Do class weights, weight decay, and label smoothing help?

RQ3 (zero-shot / few-shot LLM with Qwen2.5) and the full tokenizer / other-model comparisons (E8–E10) remain for follow-up work after the deadline.

2. Dataset and preprocessing

Stage 1 uses the public **Urdu News Dataset 1M** (Kaggle copy of Mendeley v3, DOI 10.17632/834vsxnb99.3). The authors allow non-commercial research use with credit to news sources. Direct scraping of several Urdu news sites was blocked by terms of use, so the released research corpus was used instead.

Item	Value
Rows kept after audit	111,860
Stage 1 labels	sports, business, entertainment, technology
sports / entertainment / business / technology	44,829 / 34,901 / 24,131 / 7,999
Split	80% train / 10% validation / 10% test
Seed	42
Split method	Normalized-content hash; near-duplicates stay in one split
Test headlines (sealed until finalist)	11,252

Table 1. Stage 1 corpus summary.

Headlines were normalized (whitespace and character variants), audited for empty or unmapped labels, and split before any tokenizer or model was fit. The test split stayed sealed until the finalist was declared on 20 October 2026.

3. Model and training setup

Main model: **xlm-roberta-base** (~278M parameters) with a four-class classification head. Hardware: Kaggle Tesla T4 (one chip visible; batch size 32). Max sequence length 64. Optimizer AdamW. Selection metric: validation macro-F1 only.

Finalist recipe (kept after the ladder)

Setting	Value
Strategy	Full fine-tune (not LoRA)
Peak learning rate	2e-5
Schedule	10% warmup, half-cosine over 20 epochs, then constant 2e-6
Weight decay	0.01
Loss	Unweighted cross-entropy (no label smoothing)

Setting	Value
Batch size / max length / seed	32 / 64 / 42
Epochs scored / best epoch	48 of planned 50 / epoch 46
Validation macro-F1	0.9481
Test macro-F1 (once)	0.9384
Checkpoint	HammadHARahim/urdu-headline-stage1-cosine

Table 2. Declared Stage 1 finalist configuration.

4. Experiment ladder

Each run changed one controlled variable. A setting joined the recipe only if it beat the kept validation macro-F1. Full run log: experiments/registry.csv.

ID	Experiment	Status	Val macro-F1	Notes
—	Majority / TF-IDF+LR	Done	0.1437 / 0.9230	Baselines
E2	Frozen encoder	Done	0.8538	2 epochs
E3	Full fine-tune	Done	0.9481	Cosine finalist
E4a	LoRA r=4	Done	0.9021 → 0.9395	Long run best LoRA
E4b	LoRA r=8	Done	0.9042 → 0.9377	30-epoch run
E4c	LoRA r=16	Done	0.9120	2-epoch ladder
E5	WD / label smoothing	Partly	≤0.9466	Did not beat 0.9481
E6	Warmup / cosine schedule	Done	0.9466 → 0.9481	Cosine won
E7	Class weights	Done	0.9430	Did not beat then-best
E1	Qwen zero/few-shot LLM	Remaining	—	RQ3
E8	Max length / headline+lead	Remaining	—	Input length
E9	mBERT / MuRIL	Remaining	—	Model comparison
E10	English-only BERT	Remaining	—	Tokenizer effect

Table 3. Assignment ladder coverage (E1–E10).

5. Main results

Run	Epochs scored	Best epoch	Val macro-F1
Warmup + linear decay	20	19	0.9466
Cosine + 2e-6 continuation (finalist)	48	46	0.9481
Cosine + label smoothing 0.1	60	36	0.9466
Best LoRA (r=4 + smoothing)	100	56	0.9395

Table 4. Headline full-model and best LoRA scores.

Observations

- Full fine-tuning clearly beat LoRA on validation macro-F1 (0.9481 vs 0.9395).
- Warmup helped; switching decay from linear to cosine + low-LR continuation gained another +0.0015.

- Label smoothing, stronger weight decay, and class weights did not beat the kept full-model score.
- Technology is the smallest class and remains the hardest label on the test set.

6. Final test evaluation (once)

On 20 October 2026 the kept cosine checkpoint was declared the Stage 1 finalist. A fresh retrain from xlm-roberta-base with only winning settings was preferred by project ADR-001 but was skipped because Kaggle GPU quota was exhausted (0.00h). The sealed test split was opened once on that checkpoint.

Metric	Score
Test macro-F1	0.9384
Test accuracy	0.9579
Test size	11,252 headlines
Eval device	CPU

Table 5. Finalist test summary.

Label	Precision	Recall	F1	Support
sports	0.9778	0.9810	0.9794	4531
business	0.9618	0.9754	0.9686	2401
entertainment	0.9533	0.9360	0.9446	3531
technology	0.8532	0.8695	0.8613	789

Table 6. Per-label test metrics.

	pred sports	pred business	pred entertainment	pred technology
true sports	4445	11	70	5
true business	10	2342	27	22
true entertainment	83	52	3305	91
true technology	8	30	65	686

Table 7. Confusion matrix (rows = true, columns = predicted).

7. Limitations and remaining work

- Stage 1 uses four labels only; politics, health, and world news need a second licensed corpus.
- E1 (LLM prompts), E8 (input length), E9 (other multilingual model), and E10 (English-only model) were not completed before the deadline.
- The preferred fresh finalist retrain from base was not run (GPU quota).
- Label noise from section-based labelling in the source corpus was not re-audited with a 50-headline manual sample in this delivery.
- Track A (Urdu tokenizer and transformer trained from scratch) is future work.

8. Conclusion

For Stage 1 Track B we delivered a reproducible fine-tuning ladder on xlm-roberta-base, selected the cosine-schedule full fine-tune on validation macro-F1 (0.9481), and scored the sealed test once (macro-F1 0.9384). LoRA reduced trainable parameters but did not match full fine-tuning. Schedule choice mattered more than label smoothing or class weights on this corpus. Code, experiment registry, and reports are published at the project repository.

9. Roles and packaging

Member	Primary focus for this delivery
Hammad Rahim	Training ladder, Kaggle runs, finalist selection, repository
Sultan Tufail	Data audit / split support, experiment logging, review
Adil Abdul Haq	Evaluation, report packaging, remaining-experiment notes

Table 8. Group roles for Stage 1 delivery.

Artifacts: GitHub repository (public), experiments/registry.csv, data/processed/*.txt score reports, private Hub checkpoint HammadHARahim/urdu-headline-stage1-cosine, and notebooks under notebooks/.